

Employee Salary Analysis & Salary Prediction Using Machine Learning

Project Report

Submitted By

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Course: B.Tech – Artificial Intelligence & Data Science

Organization: Syntecxhub Internship

Task: Task 4 – End-to-End Capstone Project

Project Overview

The objective of this project was to analyze employee salary data and build a machine learning model to predict whether an employee earns more than or less than **\$50K per year**. The project demonstrates the complete data analytics workflow, including data preprocessing, exploratory data analysis (EDA), visualization, machine learning, and model evaluation.

The **Adult Census Income Dataset** was used for this project. Python and its data science libraries were utilized to clean the data, generate meaningful insights, and train a predictive model.

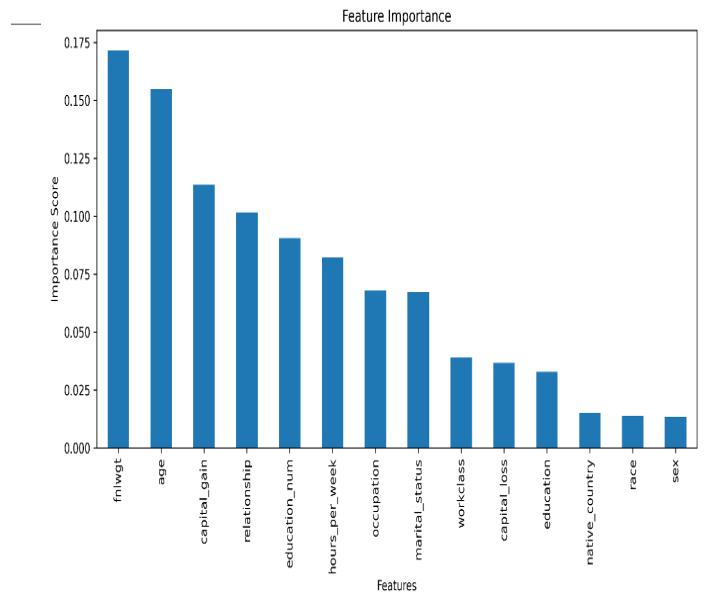
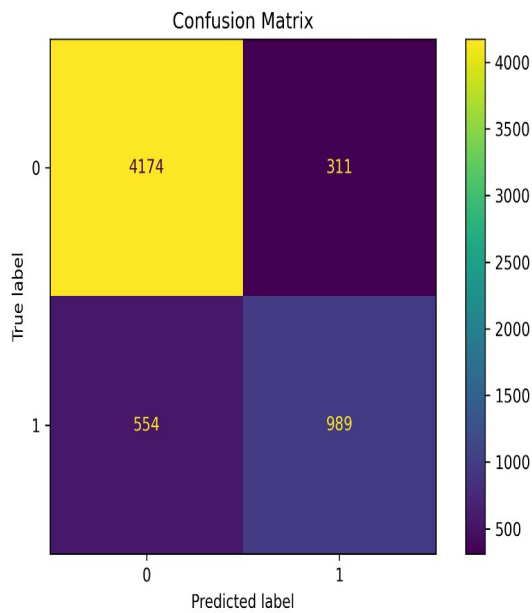
Tools & Technologies

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Scikit-learn
 - Jupyter Notebook
 - Git & GitHub
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Project Workflow

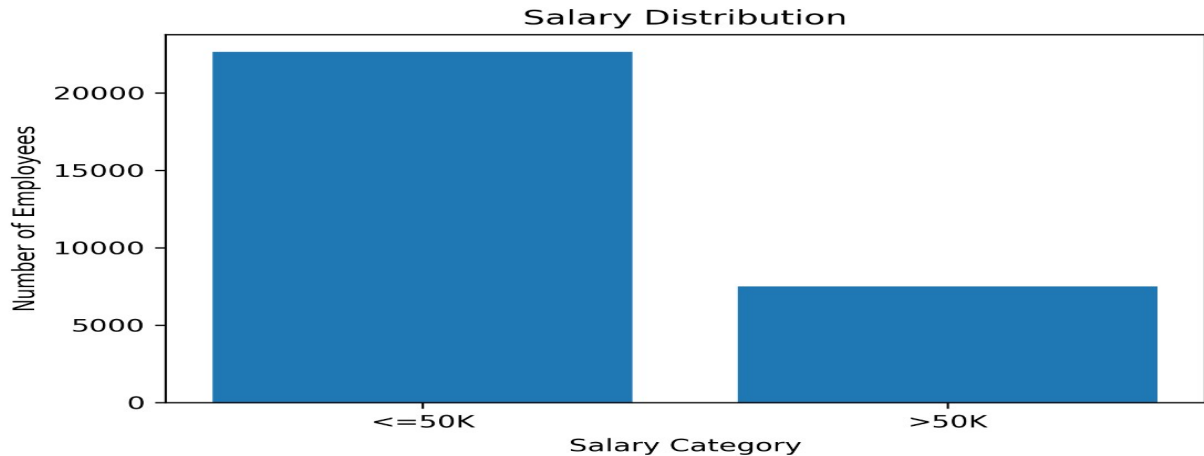
The project was completed in the following stages:

- Loaded and explored the dataset.
- Cleaned the data by handling missing values and removing duplicate records.
- Performed Exploratory Data Analysis (EDA) to understand salary trends.
- Created multiple visualizations, including salary distribution, age distribution, education analysis, occupation analysis, gender analysis, and correlation heatmap.
- Encoded categorical features using Label Encoding.
- Split the dataset into training and testing sets.
- Trained a **Random Forest Classifier** for salary prediction.
- Evaluated the model using Accuracy Score, Classification Report, and Confusion Matrix.
- Analyzed feature importance to identify the factors that most influence salary prediction.



Key Findings

- Most employees in the dataset earn $\leq 50K$ annually.
- Employees with higher education levels are more likely to earn higher salaries.
- Occupation and working hours significantly influence salary.
- Capital Gain is one of the strongest predictors of higher income.
- The Random Forest model successfully classified salary categories with good predictive performance.



Conclusion

This project successfully demonstrated an end-to-end data analytics and machine learning pipeline. It enhanced practical skills in data cleaning, visualization, predictive modeling, and model evaluation using Python. The project also provided valuable business insights that can support salary analysis and decision-making. This capstone project reflects real-world data science practices and serves as a strong addition to my academic portfolio and GitHub profile.

Project Repository

GitHub:

<https://github.com/alishamansoori004-sketch/Employee-Salary-Analysis>

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